▸ MANUEL EDUARDO TAPIA-NAVARRO, *From semantic minimalism to mathematical minimalism.*
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A non-classical mathematical theory is one whose underlying logic is incompatible with classical logic and is therefore weaker. This gives rise to a perplexity: certain statements formulated within a non-classical theory do not have the same meaning as typographically identical statements formulated within its classical counterpart. A simple example is presented by Thomas [2]. The formula $\forall x \exists y \forall z (z \in y \leftrightarrow z = x)$ classically defines singleton sets, but in a theory based on **LP** has a different meaning. The conclusion appears to be that non-classical theories may concern entities entirely different from those of their classical counterparts.

In computability theory, Martinez-Aviña [1] argued that the concept of computability does not depend on logic, demonstrating this by exhibiting two distinct theories of computability in which the concept remains identical. He considers the case of the effective topos, and dualizes its internal logic, thereby producing a logic that is tolerant to inconsistencies. This strategy relies on the fact that the natural number object has the same universal property in both the effective topos and the complement effective topos.

In this talk, I argue that Martinez-Aviña's strategy can be applied even when the theory is not formalized in the language of category theory, provided that the objects in the theory exhibit the same universal properties as in category theory and that the underlying logic satisfies certain minimal conditions.

[1] Martínez-Aviña, F., *Changing the logic without changing the subject: the case of computability*, ***Journal of Logic and Computation***, vol. 35 (2025), no. 2.

[2] Thomas, M., *Expressive Limitations of Naive Set Theory in LP and Minimally Inconsistent LP*, ***The Review of Symbolic Logic***, vol. 7 (2014), no. 2, pp. 341–350.